

Muhammad Achad Ansori

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R&D Engineer with hands-on experience across robotics, embedded systems, and industrial automation. Proven track record in designing and deploying electromechanical systems — from marine robots and offshore ROV operations to demolition robotics and smart manufacturing. Driven to bridge hardware and software to build intelligent systems that solve real-world industrial challenges.

Education

Bachelor of Applied Science, Electronics Engineering

Aug 2021 - Aug 2025

Politeknik Elektronika Negeri Surabaya (PENS)

Surabaya, Indonesia

- Advisor: Arif Irwansyah
- Thesis: "Autonomous Surface Vehicle with Collision Avoidance for Offshore Energy Exploration"

Machine Learning Cohort, Bangkit Academy

Google, Tokopedia, Gojek, Traveloka

Aug 2024 - Dec 2024

- Completed 5-month intensive program in Machine Learning, Deep Learning, TensorFlow, and Computer Vision
- Led capstone project "InSight" — YOLO-based defect detection system for SME quality control on Jetson Nano

Professional Appointments

Research and Development Engineer • Contract

PT Aldzama

Feb 2026 - Present

- Researching and developing robotic systems for industrial demolition applications
- Designing embedded control architectures and system integration for heavy-duty robotic platforms
- Collaborating with cross-functional teams to transition prototypes into field-ready solutions

Demolition Robot Technician • Contract

PT Aldzama

Sep 2025 - Jan 2026

- Developed a custom control unit for demolition robots from scratch, significantly reducing dependency on expensive OEM replacement parts
- Reverse-engineered and modified the original remote control system to integrate with the in-house built controller
- Promoted to R&D Engineer role after 3 months of controller development based on demonstrated engineering capability

Electronics Engineer • Internship

PT Venambak Kail Dipantara

Dec 2024 - May 2025

- Redesigned PCB layout with ESP32 for FeederVE control box, handled full production cycle from assembly to panel box integration for 50+ units
- Developed embedded software and electronic systems for AutoVE (autonomous mobile aerator) using Python on a low-cost mini PC
- Resolved electromagnetic interference issues on BoxVE system through PCB redesign, ensuring stable operation

ROV Pilot and Engineer • Collaborative Project

PT Pertamina Hulu Energi West Madura Offshore

Jul 2024 - Aug 2024

- Operated ROVs for underwater hull inspections at offshore RIG Taurus Madura, inspecting 3 RIG points at 20m depth over a 1-week survey period
- Performed maintenance and troubleshooting on ROV control systems, cameras, and sensors
- Collected and analyzed visual/sensor data for hull condition assessment, compiling technical reports for maintenance decisions

Maintenance and Improvement Engineer • Internship

PT Toshin Prima Fine Blanking (Indoprima Group) Feb 2024 - Jul 2024

- Developed and deployed YOLOv4-based visual inspection system on Jetson Nano achieving ~29 FPS for real-time OK/NG quality control detection
- Built IoT monitoring system for 2 critical factory chiller units using Raspberry Pi with web dashboard, enabling real-time monitoring to prevent factory-wide downtime
- Performed routine maintenance and repair of industrial fine blanking machines

Embedded Developer

Penship EEPIS Marine Robotics Team

Mar 2023 - Nov 2024

- Built entire electrical and software systems from zero, progressing from Python+MAVLink to ROS-based autonomy stack
- Developed computer vision pipelines progressing from HSV-based detection to deep learning (YOLO) for autonomous navigation on AUV and ASV platforms
- Achieved 3rd Place National in KRBAI and 2nd Place Regional, with multiple finalist appearances in national competitions

General Manager

Penship EEPIS Marine Robotics Team

Mar 2023 - Dec 2023

- Led 15-member team with 4 robot platforms, establishing weekly meeting culture from scratch
- Aligned hardware, software, and mechanical workstreams to competition deadlines

Hardware Specialist

Penship EEPIS Marine Robotics Team

Mar 2022 - Feb 2023

- Engineered electrical systems for Autonomous Semi Submarine Vehicle — propulsion, power management, and waterproofing
- Designed custom PCBs aligned with mechanical specifications for underwater operation
- Supported software team with microcontroller programming for mission-based autonomous movements

Awards and Recognitions

2nd Place, Final Project Competition — Applied Bachelor of Electronics Engineering, PENS	2025
3rd Place National, Underwater Robotics Competition (KRBAI) — Kontes Robot Indonesia	2023
2nd Place Region 2, Underwater Robotics Competition (KRBAI) — Kontes Robot Indonesia	2023
Funded Proposal, Pekan Kreativitas Mahasiswa (PKM) — Ministry of Education	2024
Most Active and Best Idea, Kaizen Event — PT Aldzama	2026

Certifications

Sea Survival and Offshore Emergency Procedure — PT Global Saftindo (Credential: GS-ADM-024-3594)	Jul 2024 - Jul 2027
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Technical Skills

- Programming:** Python, C/C++, JavaScript, Bash
- Embedded Systems:** STM32, ESP32, Arduino, Raspberry Pi, Jetson Nano, MAVLink, PID Control, UART/SPI/I2C, RTOS
- Robotics & Autonomy:** ROS, OpenCV, YOLO, TensorFlow, PyTorch, Sensor Integration, Navigation, Linux
- Electrical:** PCB Design, Soldering & Prototyping, Wiring & Panel Assembly, Motor Drivers, Power Distribution, Oscilloscope & Multimeter
- Internet of Things:** IoT Monitoring Systems, Web Dashboard, MQTT, Flutter, React, Git
- Languages:** Bahasa Indonesia (Native), English (Professional Working)